

14	The piston is pushed very slowly into a metal cylinder containing an ideal gas. If the temperature of the gas does not change in the process. Which of the following statements is <i>incorrect</i>?
A	The work done on the gas is equal to the heat lost by the gas.
B	The pressure of the gas increases.
C	The average kinetic energy of the gas molecules increases due to the work done on the gas.
D	The internal energy of the gas remains constant.
	Answer: C Metal → good conductor of heat. There can be heat transfer in/out of the cylinder. A: $\Delta U = \frac{3}{2}nR\Delta T = 0$. Hence by First Law of thermodynamics, $W = -Q$. Correct.

B: Ideal gas equation, $pV = nRT = \text{constant}$ since T & n are both constants. So since V decreases, p increases. Correct

C: $\langle E_K \rangle = \frac{3}{2} kT$. Same temperature means that the mean translational KE is the same.
Incorrect

D: internal energy of an ideal gas = $\frac{3}{2} nRT$. For a fixed n , internal energy remains constant at constant T .